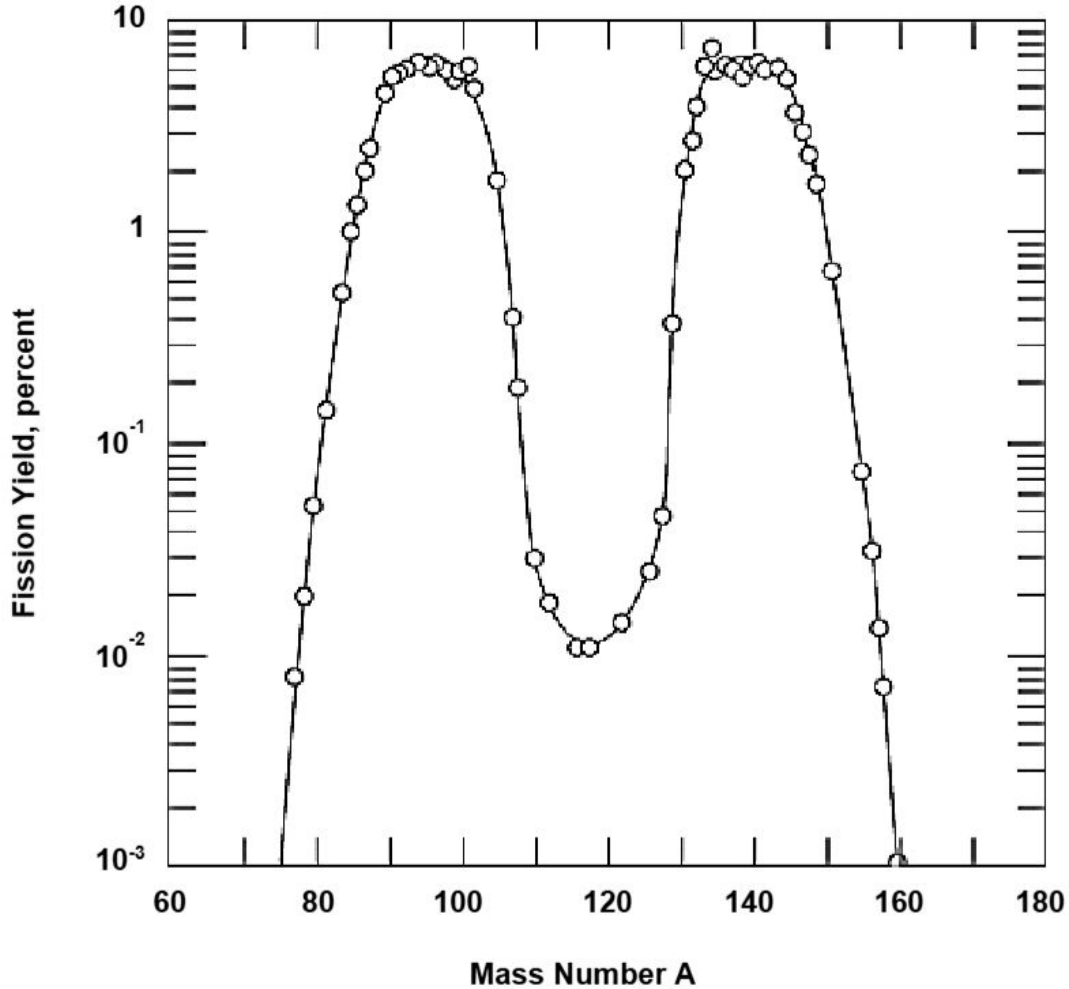
7	When a very strong earthquake of magnitude 9.0 hit Japan on 11 March 2011, the accompanying tsunami caused the devastation of the Fukushima Dai-ichi Nuclear Power Plant (Fk-1). A large amount of radioactive material leaked out of Fk-1 as a result, mainly through cracks made by the quake on some reactors, explosions, intentional vents to relieve pressure, and leakage of cooling water that was contaminated with the melted fuel rod debris. Fk-1 was capable of generating up to 4.7 gigawatts of power through neutron-induced fissions of Uranium-235 (^{235}U) nuclei. Fission resulting from neutron absorption is called *induced* fission. Some nuclides can also undergo *spontaneous* fission, but that is quite rare. Nuclear fission is a decay process in which an unstable nucleus splits into two fragments of comparable mass, although nearly equal mass is unlikely. Over 100 different nuclides, representing more than 20 different elements, have been found among the fission products. Figure 7.1 shows the distribution of mass numbers for fission fragments from the fission of ^{235}U. Fission yield refers to the percentage of a nuclide of a fission fragment produced per fission. Two or three free neutrons usually appear along with the fission fragments.



Source: https://www.quora.com/Why-is-promethium-produced-from-uranium-235

Figure 7.1

The induced fission of ^{235}U may be represented by a nuclear equation of the form

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{a}_{b}P + {}^{c}_{d}Q + (2 \text{ or } 3){}^{1}_{0}n + \text{energy}$$

where P represents the lower mass number fission fragment, and Q represents the higher mass number fission fragment.

One day after the earthquake, elevated levels of Iodine-131, Cesium-134 and Cesium-137, which are common fission fragments of Uranium-235, were detected near the nuclear plant site. Figure 7.2 shows some general information of the three fission fragments.

Isotope	Symbol	Half-life
Iodine-131	$^{131}_{53}I$	8.0197 days
Cesium-134	$^{134}_{55}Cs$	2.065 years
Cesium-137	$^{137}_{55}Cs$	30.17 years

Figure 7.2

(a) Explain the difference between *induced* and *spontaneous* fission.

.....

		.....	
		.....	[1]
	L1	Unlike spontaneous fission, **induced fission requires** (initial) **neutron absorption to occur**. *Candidates need to answer this question by reading the passage in context.*	**B1**
	(b)	Suggest why the *percentage fission yield* in Figure 7.1 is shown on a logarithmic scale.	
		.....	
		.....	
		.....	[1]
	L3	Percentage fission yield varies over a wide range of many orders of magnitude. OR Logarithmic scale compresses the scale to accommodate the very wide range of percentage fission yield values. OR To enable plots on the graph to be well spaced while representing the very wide range of percentage fission yield values.	**B1**
	(c)	Using Figure 7.1, state the two most common ranges of mass numbers of the fission fragments (P and Q) of ^{235}U.	
		most common range of mass numbers of P:	[1]
		most common range of mass numbers of Q:	[1]
	L1	From Figure 7.1, Most common range of mass numbers of **P**: **90 to 100 (Accept lower boundary value 88 to 90, and, upper boundary value 100 to 102.)** Most common range of mass numbers of **Q**: **132 to 142 (Accept lower boundary value 132 to 134, and, upper boundary value 142 to 145.)** Note that Question has already defined P as the fragment with *lower* mass. So answers for P and Q cannot swop. Marking points: - Correct reading of the ranges of values. - Answers for P and Q not swopped.	**A1** **A1**
	(d)	**(i)** Uranium-235 undergoes nuclear fission to produce Rubidium-97 and Cesium-137. Write down the nuclear equation to represent this reaction. Use the chemical symbol Rb to represent Rubidium.	
		.....	[2]
	L2	$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{97}_{37}Rb + {}^{137}_{55}Cs + 2\,{}^{1}_{0}n$ • Correct value of proton number of Rb (= 37). • Deduced that 2 neutrons are emitted. • Deduct 1 mark for other mistakes e.g. used "=" instead of "→"; did not include $^{1}_{0}n$ on the LHS of the equation; wrong notations e.g. n^{1}_{0} or $^{2}_{0}n$	 **A1** **A1**

			instead of $2\,{}^{1}_{0}n$ Note: General equation is given in the question, hence no mark purely for writing the equation.	
		(ii)	With reference to Figure 7.1, state the percentage fission yield of Cesium-137 and Rubidium-97.	
			percentage yield of Cesium-137= .....%	
			percentage yield of Rubidium-97 = ...%	[1]
		L1	6% for both. (Accept also 6.5%.) Mark for correct method (clear indication of points read on Figure 7.1). Accept also if answers given are 6.5% for one, and 6% for the other. *Further interesting note on Figure 7.1:* *By conservation of mass number before and after the fission, it is necessary that the two fission fragments are either read off the two inner parts of the "M" shaped curve in Figure 7.1, or the two outer parts.*	**A1**
		(iii)	Suggest why both the answers in **(d)(ii)** are similar.	
			.....	
			.....	
			.....	[1]
		L3	Percentage yields of both should be similar since **they are produced from the same (unique) fission process**. So the probabilities of both of them forming are equal/same. *Do NOT accept:* "In the nuclear equation, Cesium-137 and Rubidium-97 are produced in a 1:1 ratio." ➔ *This is NOT accepted as an answer because: Firstly, it is rephrasing the question which already specifies that the answers are 'similar'. Secondly, it does not address the meaning of 'percentage yield' which the values in (d)(ii) refer to. In paragraph 2 of the given text, the concept of percentage yield comes in because U-235 can undergo many different fission processes, i.e. it does not necessarily split into Cs and Rb. There is also a need to justify that Cs and Rb are produced in the same unique fission process, i.e. no other fission possibilities of U-235 can yield either one of them without the other.* *Do NOT accept:* "due to conservation of mass number", *because question asks for reasons for similarity in the percentage fission yield, not about deducing 1 product from the other given product. Moreover, there are neutrons not accounted for by this explanation.*	**B1**
		(iv)	Determine the **two** possible mass numbers of the other fission fragment produced if one of the fragments has a mass number of 81 in the nuclear fission of Uranium-235.	

			mass number = ... or	[2]
		L2	Using $^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{81}_{b}P + {}^{c}_{d}Q + (2 \text{ or } 3)\,{}^{1}_{0}n + \text{energy}$ By conservation of mass number, **235 + 1 = 81 + c + (2 or 3)** **c = 153 or 152**	**A2**
	(e)	Determine which isotope in Figure 7.2 will be about 47% as radioactive after 33 years. Show your reasoning.		
			isotope:	[2]
	L2	Method 1: **47% ≈ 50%, hence** **33 years is close to the half-life of the isotope.** From Figure 7.2, the isotope with a half-life close to 33 years is **Cesium-137**. Method 2: **Activity at time t, $A = A_o e^{-\lambda t}$ where $\lambda = \ln 2 / t_{1/2}$** **"47% as radioactive" → $A / A_o = 0.47$** **$0.47 = e^{-\lambda(33\text{years})}$** $\lambda = 0.022879\ \text{years}^{-1}$ $\ln 2 / t_{1/2} = 0.022879\ \text{years}^{-1}$ **$t_{1/2}$ = ~30 years** **Answer: Cesium-137**	**M1** **A1** **[M1]** **[A1]**
	(f)	The release of Iodine-131, Cesium-134 and Cesium-137 into the air, ground and seawater were monitored closely by the Japanese authority because they readily decay by emitting beta-particles and gamma-rays. For example, Cesium-137 decays into Barium-137 (Ba) through beta-emission. 95% of Cesium-137 decays to a metastable excited nuclear energy state (Barium-137m) and is responsible for all emission of gamma-rays, while the other 5% directly decays to the ground state (Barium-137) of the nucleus. Figure 7.4 illustrates the decay scheme of Cesium-137.		

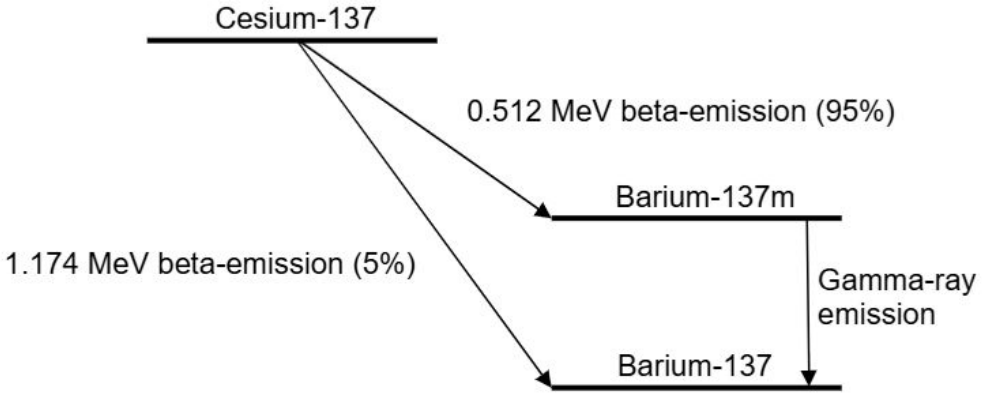


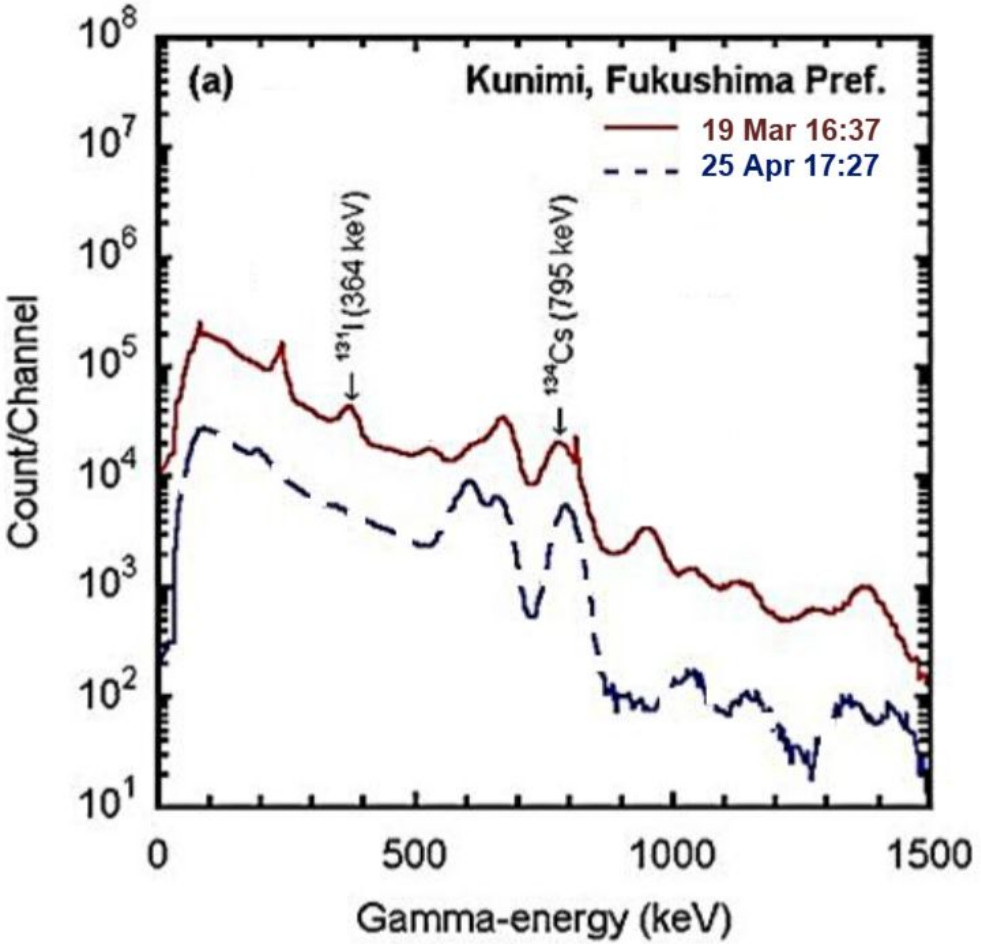
Figure 7.4

Determine the photon energy of the gamma-rays emitted during Cesium-137 decay.

energy = ...MeV [2]

	L1	Energy of gamma-ray photon **= 1.174 – 0.512** **= 0.662 MeV**	**M1** **A1**

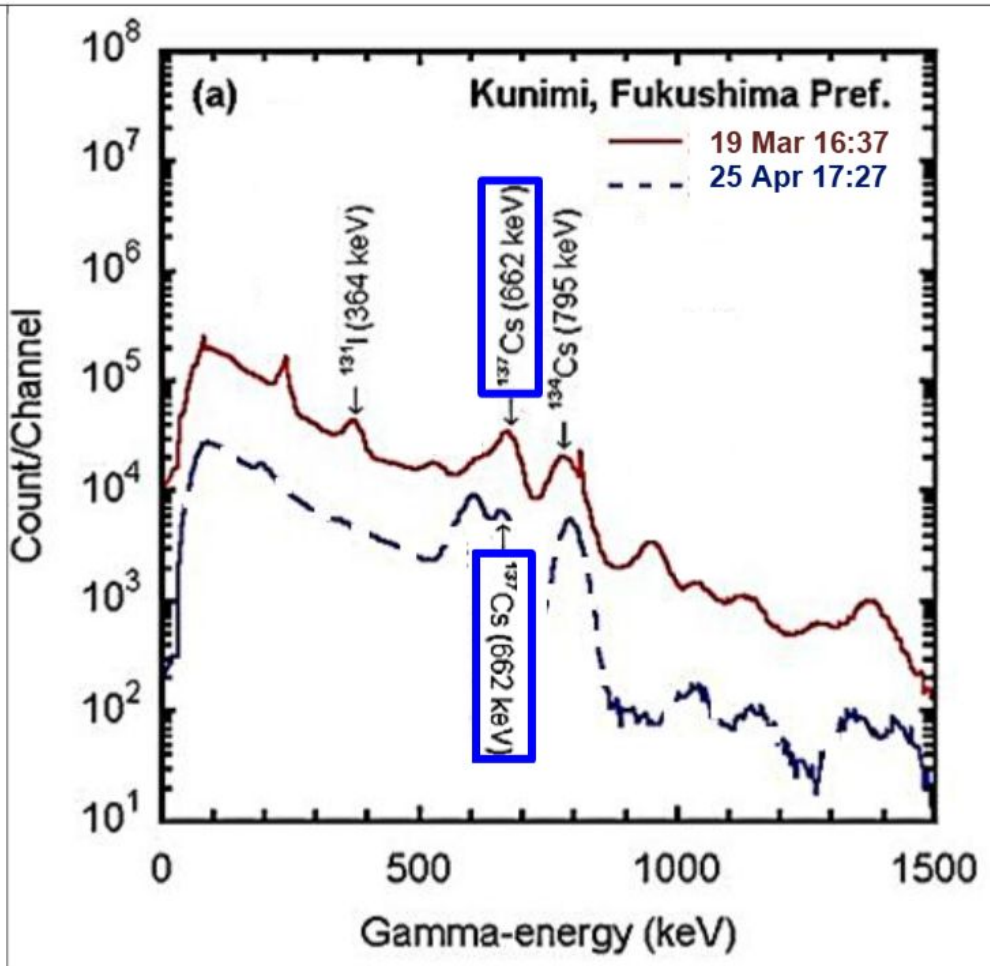
(g) Gamma spectroscopy is a technique used by scientists to identify specific radioactive products from the many possible products that can be produced from the nuclear fission of Uranium-235. Figure 7.5 shows the gamma spectra measured at Kunimi, Fukushima Prefecture on both 19 March 2011 and 25 April 2011. Photon peaks generated by Tellurium-132, Iodine-131, -132, and Cesium-134, -136 and -137 were observed in the spectra given in Figure 7.5. The photon peaks corresponding to Iodine-131 and Cesium-134 are indicated in Figure 7.5.



Source: https://www.nature.com/articles/srep00087 (modified)

Figure 7.5

		(i)	Using your answer in **(f)**, draw an arrow on Figure 7.5 to show the position of Cesium-137 in the gamma spectra. Label this arrow Cs-137.	[1]
		L2	**0.662 MeV = 662 keV** ***Accept indication on either one of the solid or dotted line graphs.*** ***Allow ECF according to candidate's answer in (f). Check for correct units conversion from MeV to keV.***	**A1**



(ii) Suggest why no photon peak of Iodine-131 was observed on 25 April 2011.

.....

.....

..... [1]

L3 As seen from Figure 7.2, Iodine-131 has a short half-life of about 8 days, hence **a month after 19 March, on 25 April, about four half-lives had passed, hence most of the iodine nuclei would have already decayed**. **B1**

OR

Half life of Iodine-131 is about 8 days, so **by 25 April, yield of Iodine-131 is only 4.42% of that on 19 March which is significantly low.**

[About 36 days passed → $\frac{t}{t_{\frac{1}{2}}} = \frac{36}{8} = 4.5\ half\ lives\ passed$ **→**

$\frac{A}{A_o} = \left(\frac{1}{2}\right)^{4.5} = 0.0442$ **= 4.42%]**

(h) Compare the three types of radioactive decay: alpha-decay, beta-decay and gamma-decay. Explain which of the three poses the greatest health risk if the source is

(i) outside the body,

.....

.....

			.....	
			.....	
			.....	[2]
		L2	**Gamma-decay. Because gamma-ray is the <u>most</u> penetrating hence <u>penetrate most deeply</u> into the body and <u>reach living tissues and vital organs.</u>** **Alpha-particle <u>cannot</u> penetrate the skin. Beta-particle <u>can penetrate the skin but cannot penetrate as deeply to reach vital organs</u>.** ***Must link to health risk for the 2nd mark.*** ***No mark for simply stating Gamma-decay without the correct reason.*** ***Comparative language is required e.g. "most penetrating"/ "highest penetrating ability".*** ***Elaboration of what kind of health problems is <u>not</u> required for part i, since the link for the 2nd mark should be made to the implication of "highest <u>penetrating</u> ability" to health → <u>greatest potential of reaching and affected more living tissues in the body.</u>*** ***(All 3 types of radiations damage cells via ionization. The danger of this ionizing ability is elaborated in part ii answer. Focus on part i is on the penetration ability.)***	**B1** **B1**
		(ii)	ingested and entered the body.	
			.....	
			.....	
			.....	
			.....	
			.....	[2]
		L2	**Alpha-decay. Because alpha-particle is the <u>most</u> ionizing hence produces the <u>most</u> number of free radicals per unit distance traveled inside the body / causes the <u>most</u> extensive mutation of cells or DNA / <u>highest</u> cancer risk.** ***Must link to health risk for the 2nd mark.*** ***No mark for simply stating alpha-decay without the correct reason.*** ****Knowledge of health risks is in the A-level syllabus.*** ***Comparative language is required e.g. "most ionizing"/ "highest ionising ability".*** ***Elaboration of what kind of health problems <u>is required</u> for part ii, since the link for the 2nd mark should be made to the implication of "highest <u>ionising</u> ability" to health → e.g. production free radicals, mutation of cells or DNA, cancer risk.*** ***Repetitive answers are not accepted e.g. can ionize cells → repeating "ionizing ability", need to elaborate on health impact.***	**B1** **B1**

[Total: 20]